Unit-III: Network Layer

❖ Design Issues in Network Layer

The Network layer is majorly focused on getting packets from the source to the destination, routing error handling, and congestion control. Before learning about design issues in the network layer, let's learn about its various functions.

- **Addressing:** Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in the network.
- **Packeting:** This is performed by Internet Protocol. The network layer converts the packets from its upper layer.
- **Routing:** It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.
- **Inter-networking:** It works to deliver a logical connection across multiple devices.

Network Layer Design Issues

The network layer comes with some design issues that are described as follows:

1. Store and Forward packet switching

The host sends the packet to the nearest router. This packet is stored there until it has fully arrived once the link is fully processed by verifying the checksum then it is forwarded to the next router till it reaches the destination. This mechanism is called "Store and Forward packet switching."

2. Services provided to the Transport Layer

Through the network/transport layer interface, the network layer transfers its **patterns** services to the transport layer. These services are described below. But before providing these services to the transfer layer, the following goals must be kept in mind

- Offering services must not depend on router technology.
- The transport layer needs to be protected from the type, number, and topology of the available router.
- The network addresses for the transport layer should use uniform numbering patterns**,** also at LAN and WAN connections.

Based on the connections there are 2 types of services provided

- **Connectionless –** The routing and insertion of packets into the subnet are done individually. No added setup is required.
- **Connection-Oriented –** Subnet must offer reliable service and all the packets must be transmitted over a single route.

3. Implementation of Connectionless Service

Packets are termed as "datagrams" and corresponding subnets as "datagram subnets". When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to the router via. a few protocols. Each data packet has a destination address and is routed independently irrespective of the packets.

4. Implementation of Connection-Oriented service:

To use a connection-oriented service, first, we establish a connection, use it, and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender. It can be done in either two ways

- **Circuit Switched Connection –** A dedicated physical path or a circuit is established between the communicating nodes and then the data stream is transferred.
- **Virtual Circuit Switched Connection –** The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.

Store-and-forward Packet Switching

Store-and-forward switching is a method of switching data packets by the switching device that receives the data frame and then checks for errors before forwarding the packets. It supports the efficient transmission of non-corrupted frames. It is generally used in telecommunication networks.

❖ Connection-Oriented Service

Connection-Oriented Service is basically a technique that is typically used to transport and send data at session layer. The data streams or packets are transferred or delivered to receiver in a similar order in which they have seen transferred by sender. It is actually a data transfer method among two devices or computers in a different network, that is designed and developed after telephone system. Whenever a network implements this service, it sends or transfers data or message from sender or source to receiver or destination in correct order and manner.
This connection service is generally provided by protocols of both network layer (signifies different path for various data packets that belongs to same message) as well as transport layer (use to exhibits independence among packets rather than different paths that various packets belong to same message will follow).

Operations :
There is a sequence of operations that are needed to b followed by users. These operations are given below:

1. **Establishing Connection –**
   It generally requires a session connection to be established just before any data is transported or sent with a direct physical connection among sessions.
2. **Transferring Data or Message –**
   When this session connection is established, then we transfer or send message or data.
3. **Releasing the Connection –**
   After sending or transferring data, we release connection.

Different Ways :
There are two ways in which connection-oriented services can be done. These ways are given below:

1. **Circuit-Switched Connection –**
   Circuit-switching networks or connections are generally known as connection-oriented networks. In this connection, a dedicated route is being established among sender and receiver, and whole data or message is sent through it. A dedicated physical route or a path or a circuit is established among all communication nodes, and after that, data stream or message is sent or transferred.
2. **Virtual Circuit-Switched Connection –**
   Virtual Circuit-Switched Connection or Virtual Circuit Switching is also known as Connection-Oriented Switching. In this connection, a preplanned route or path is established before data or messages are transferred or sent. The message Is transferred over this network is such a way that it seems to user that there is a dedicated route or path from source or sender to destination or receiver.

Types of Connection-Oriented Service:

Service	Example
Reliable Message Stream	Sequence of pages, etc.
Reliable Byte Stream	Song Download, etc.
Unreliable Connection	VoIP (Voice Over Internet Protocol)

Advantages:

- It kindly support for quality of service is an easy way.
- This connection is more reliable than connectionless service.
- Long and large messages can be divided into various smaller messages so that it can fit inside packets.
- Problems or issues that are related to duplicate data packets are made less severe.

Disadvantages:

- In this connection, cost is fixed no matter how traffic is.
- It is necessary to have resource allocation before communication.
- If any route or path failures or network congestions arise, there is no alternative way available to continue communication.

❖ Connection-less Service

A **Connectionless Service** is technique that is used in data communications to send or transfer data or message at Layer 4 i.e., Transport Layer of Open System Interconnection model. This service does not require session connection among sender or source and receiver or destination. Sender starts transferring or sending data or messages to destination.

In other words, we can say that connectionless service simply means that node can transfer or send data packets or messages to its receiver even without session connection to receiver. Message is sent or transferred without prior arrangement. This usually works due to error handling protocols that allow and give permission for correction of errors just like requesting retransmission.

In this service, network sends each packet of data to sender one at a time, independently of other packets. But network does not have any state information to determine or identify whether packet is part of stream of other packets. Even the network doesn't have any knowledge and information about amount of traffic that will be transferred by user. In this, each of data packets has source or destination address and is routed independently from source to destination.

Therefore, data packets or messages might follow different paths to reach destination. Data packets are also called datagrams. It is also similar to that of postal services, as it also carries full address of destination where message is to send. Data is also sent in one direction from source to destination without checking that destination is still present there or not or if receiver or destination is prepared to accept message.

Connectionless Protocols :
These protocols simply allow data to be transferred without any link among processes. Some of data packets may also be lost during transmission. Some of protocols for connectionless services are given below:

- **Internet Protocol (IP)**
  This protocol is connectionless. In this protocol, all packets in IP network are routed independently. They might not go through same route.
- **User Datagram Protocol (UDP)**
  This protocol does not establish any connection before transferring data. It just sends data that's why UDP is known as connectionless.
- **Internet Control Message Protocol (ICMP)**
  ICMP is called connectionless simply because it does not need any hosts to handshake before establishing any connection.
- **Internetwork Packet Exchange (IPX)**
  IPX is called connectionless as it doesn't need any consistent connection that is required to be maintained while data packets or messages are being transferred from one system to another.

Types of Connectionless Services:

Service	Example
Unreliable Datagram	Electronic Junk Mail, etc.
Acknowledged Datagram	Registered mail, text messages along with delivery report, etc.
Request Reply	Queries from remote databases, etc.

Advantages:

- It is very fast and also allows for multicast and broadcast operations in which similar data are transferred to various recipients in a single transmission.
- The effect of any error occurred can be reduced by implementing error-correcting within an application protocol.
- This service is very easy and simple and is also low overhead.
- At the network layer, host software is very much simpler.
- No authentication is required in this service.
- Some of the application doesn't even require sequential delivery of packets or data. Examples include packet voice, etc.

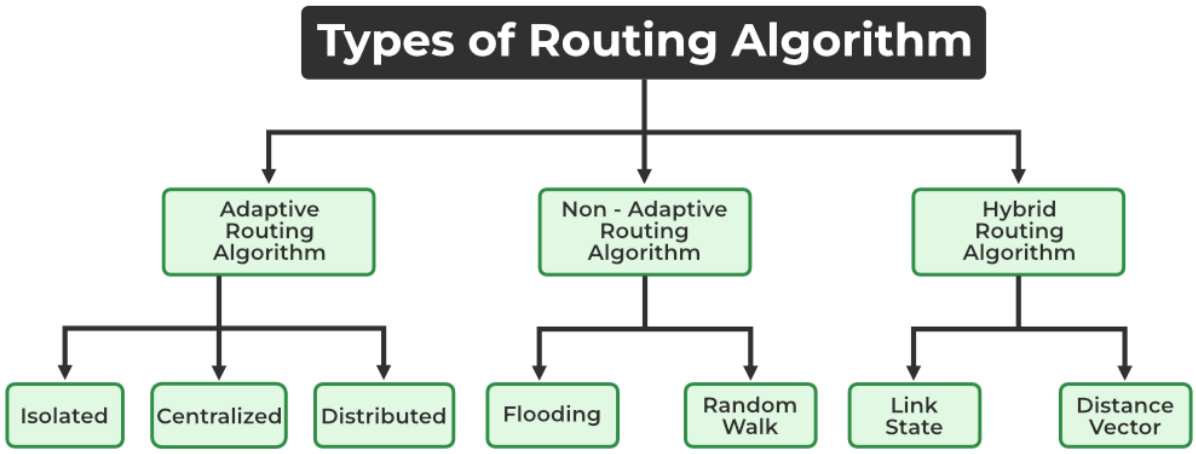
Disadvantages:

- This service is less reliable as compared to connection-oriented service.
- It does not guarantee that there will be no loss, or error occurrence, misdelivery, duplication, or out-of-sequence delivery of the packet.
- They are more prone towards network congestions.

Classification of Routing Algorithms

The routing algorithms can be classified as follows:

1. Adaptive Algorithms
2. Non-Adaptive Algorithms
3. Hybrid Algorithms



Types of Routing Algorithm

1. Adaptive Algorithms

These are the algorithms that change their routing decisions whenever network topology or traffic load changes. The changes in routing decisions are reflected in the topology as well as the traffic of the network. Also known as dynamic routing, these make use of dynamic information such as current topology, load, delay, etc. to select routes. Optimization parameters are distance, number of hops, and estimated transit time.

Further, these are classified as follows:

- **Isolated:** In this method each, node makes its routing decisions using the information it has without seeking information from other nodes. The sending nodes don't have information about the status of a particular link. The disadvantage is that packets may be sent through a congested network which may result in delay. Examples: Hot potato routing, and backward learning.

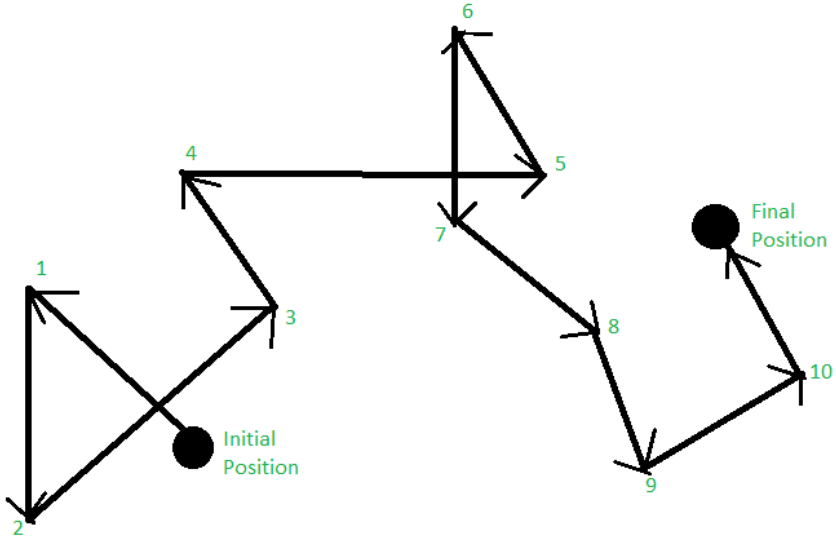
- **Centralized:** In this method, a centralized node has entire information about the network and makes all the routing decisions. The advantage of this is only one node is required to keep the information of the entire network and the disadvantage is that if the central node goes down the entire network is done. The link state algorithm is referred to as a centralized algorithm since it is aware of the cost of each link in the network.
- **Distributed:** In this method, the node receives information from its neighbors and then takes the decision about routing the packets. A disadvantage is that the packet may be delayed if there is a change in between intervals in which it receives information and sends packets. It is also known as a decentralized algorithm as it computes the least-cost path between source and destination.

2. Non-Adaptive Algorithms

These are the algorithms that do not change their routing decisions once they have been selected. This is also known as static routing as a route to be taken is computed in advance and downloaded to routers when a router is booted.

Further, these are classified as follows:

- **Flooding:** This adapts the technique in which every incoming packet is sent on every outgoing line except from which it arrived. One problem with this is that packets may go in a loop and as a result of which a node may receive duplicate packets. These problems can be overcome with the help of sequence numbers, hop count, and spanning trees.
- **Random walk:** In this method, packets are sent host by host or node by node to one of its neighbors randomly. This is a highly robust method that is usually implemented by sending packets onto the link which is least queued.



Random Walk

3. Hybrid Algorithms

As the name suggests, these algorithms are a combination of both adaptive and non-adaptive algorithms. In this approach, the network is divided into several regions, and each region uses a different algorithm.

Further, these are classified as follows:

- Link-state: In this method, each router creates a detailed and complete map of the network which is then shared with all other routers. This allows for more accurate and efficient routing decisions to be made.
- Distance vector: In this method, each router maintains a table that contains information about the distance and direction to every other node in the network. This table is then shared with other routers in the network. The disadvantage of this method is that it may lead to routing loops.

Types of Routing Protocol in Computer Networks

1. Routing information protocol (RIP)

One of the earliest protocols developed is the inner gateway protocol, or RIP. we can use it with local area networks (LANs), that are linked computers in a short range, or wide area networks (WANs), which are telecom networks that cover a big range. Hop counts are used by the Routing Information Protocol (RIP) to calculate the shortest path between networks.

2. Interior gateway protocol (IGRP)

IGRP was developed by the multinational technology corporation Cisco. It makes use of many of the core features of RIP but raises the maximum number of supported hops to 100. It might therefore function better on larger networks. IGRPs are elegant and distance-vector protocols. In order to work, IGRP requires comparisons across indicators such as load, reliability, and network capacity. Additionally, this kind updates automatically when things change, such as the route. This aids in the prevention of routing loops, which are mistakes that result in an unending data transfer cycle.

3. Exterior Gateway Protocol (EGP)
Exterior gateway protocols, such as EGP, are helpful for transferring data or information between several gateway hosts in autonomous systems. In particular, it aids in giving routers the room they need to exchange data between domains, such as the internet.
4. Enhanced interior gateway routing protocol (EIGRP)
This kind is categorised as a classless protocol, inner gateway, and distance vector routing. In order to maximise efficiency, it makes use of the diffusing update method and the dependable transport protocol. A router can use the tables of other routers to obtain information and store it for later use. Every router communicates with its neighbour when something changes so that everyone is aware of which data paths are active. It stops routers from miscommunicating with one another. The only external gateway protocol is called Border Gateway Protocol (BGP).
5. Open shortest path first (OSPF)
OSPF is an inner gateway, link state, and classless protocol that makes use of the shortest path first (SPF) algorithm to guarantee effective data transfer. Multiple databases containing topology tables and details about the network as a whole are maintained by it. The ads, which resemble reports, provide thorough explanations of the path's length and potential resource requirements. When topology changes, OSPF recalculates paths using the Dijkstra algorithm. In order to guarantee that its data is safe from modifications or network intrusions, it also employs authentication procedures. Using OSPF can be advantageous for both large and small network organisations because to its scalability features.
6. Border gateway protocol (BGP)
Another kind of outer gateway protocol that was first created to take the role of EGP is called BGP. It is also a distance vector protocol since it performs data package transfers using the best path selection technique. BGP defines communication over the internet. The internet is a vast network of interconnected autonomous systems. Every autonomous system has autonomous system number (ASN) that it receives by registering with the Internet Assigned Numbers Authority.

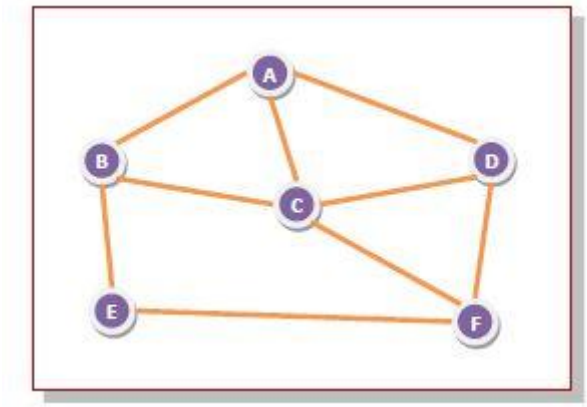
Difference between Routing and Flooding

Routing	Flooding
A routing table is required.	No Routing table is required.
May give the shortest path.	Always gives the shortest path.
Less Reliable.	More Reliable.
Traffic is less.	Traffic is high.
No duplicate packets.	Duplicate packets are present.

- **Flooding**

Flooding is a **non-adaptive routing technique** following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

For example, let us consider the **network** in the figure, having six routers that are connected through transmission lines.



Using flooding technique −

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.

Types of Flooding

Flooding may be of three types −

- **Uncontrolled flooding** − Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** − They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).
- **Selective flooding** − Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

Advantages of Flooding:

- Highly Robust, emergency or immediate messages can be sent (eg military applications)
- Set up the route in virtual circuit
- Flooding always chooses the shortest path
- Broadcast messages to all the nodes

Disadvantages of Flooding:

- Network congestion: Flooding can cause a significant amount of traffic in the network, leading to congestion. This can result in slower network speeds and delays in delivering data packets.
- Wastage of network resources: Flooding uses a lot of network resources, including bandwidth and processing power, to deliver packets. This can result in the wastage of valuable network resources and reduce the overall efficiency of the network.

- Security risks: Flooding can be used as a tool for launching various types of attacks, including denial of service (DoS) attacks. Attackers can flood the network with data packets, which can overload the network and cause it to crash.
- Inefficient use of energy: Flooding can result in an inefficient use of energy in wireless networks. Since all nodes receive every packet, even if they are not the intended recipient, they will still need to process it, which can waste energy and reduce the overall battery life of mobile devices.
- Difficulty in network troubleshooting: Flooding can make it difficult to troubleshoot network issues. Since packets are sent to all nodes, it can be challenging to isolate the cause of a problem when it arises.

❖ Shortest Path Routing

It refers to the algorithms that help to find the shortest path between a sender and receiver for routing the data packets through the network in terms of shortest distance, minimum cost, and minimum time.

- It is mainly for building a graph or subnet containing routers as nodes and edges as communication lines connecting the nodes.
- Hop count is one of the parameters that is used to measure the distance.
- **Hop count**: It is the number that indicates how many routers are covered. If the hop count is 6, there are 6 routers/nodes and the edges connecting them.
- Another metric is a geographic distance like kilometers.
- We can find the label on the arc as the function of bandwidth, average traffic, distance, communication cost, measured delay, mean queue length, etc.

❖ Distance Vector Routing (DVR) Protocol

Distance Vector Routing (DVR) Protocol is a method used by routers to find the best path for data to travel across a network. Each router keeps a table that shows the shortest distance to every other router, based on the number of hops (or steps) needed to reach them. Routers share this information with their neighbors, allowing them to update their tables and find the most efficient routes. This protocol helps ensure that data moves quickly and smoothly through the network.

Distance Vector Routing Algorithm

The protocol requires that a router inform its neighbors of topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as the Bellman-Ford algorithm).

Bellman-Ford Basics

Each router maintains a Distance Vector table containing the distance between itself and All possible destination nodes. Distances, based on a chosen metric, are computed using information from the neighbors' distance vectors.

```
Information kept by DV router:

•  Each router has an ID
•  Associated with each link connected to a router,
   there is a link cost (static or dynamic).
•  Intermediate hops

Distance Vector Table Initialization:
•  Distance to itself = 0
•  Distance to ALL other routers = infinity number.
```

Distance Vector Algorithm

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
    - It receives a distance vector from a neighbor containing different information than before.
    - It discovers that a link to a neighbor has gone down.

The DV calculation is based on minimizing the cost to each destination

```
Dx(y) = Estimate of least cost from x to y
C(x,v) =  Node x knows cost to each neighbor v
Dx   =  [Dx(y): y ? N ] = Node x maintains distance vector
Node x also maintains its neighbors' distance vectors
– For each neighbor v, x maintains Dv = [Dv(y): y ? N ]
```

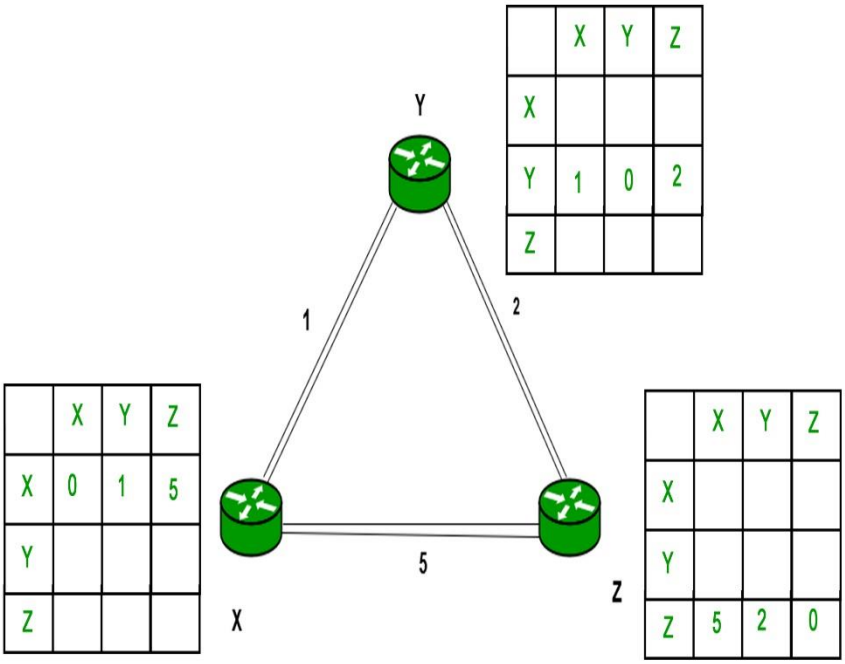
Note:

- From time-to-time, each node sends its own distance vector estimate to neighbors.
- When a node x receives new DV estimate from any neighbor v, it saves v's distance vector and it updates its own DV using B-F equation:

  ```
  Dx(y) = min { C(x,v) + Dv(y), Dx(y) } for each node y ? N
  ```

Example :

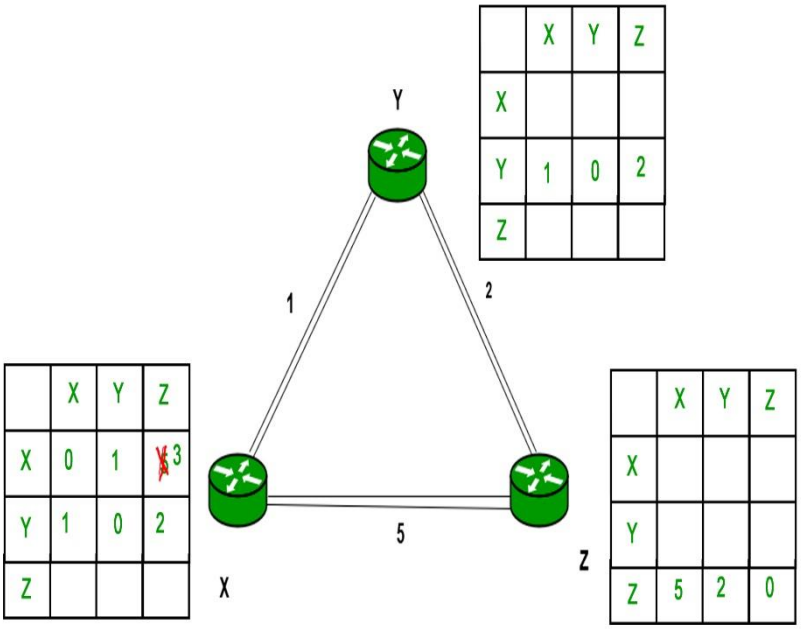
Consider 3-routers X, Y and Z as shown in figure. Each router have their routing table. Every routing table will contain distance to the destination nodes.



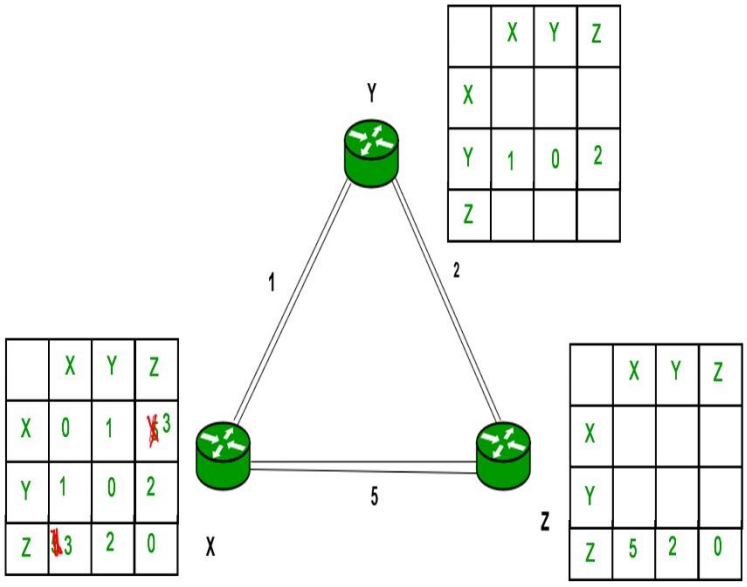
Consider router X , X will share it routing table to neighbors and neighbors will share it routing table to it to X and distance from node X to destination will be calculated using bellmen- ford equation.

```
Dx(y) = min { C(x,v) + Dv(y)} for each node y ? N
```

As we can see that distance will be less going from X to Z when Y is intermediate node(hop) so it will be update in routing table X.



Similarly for Z also –

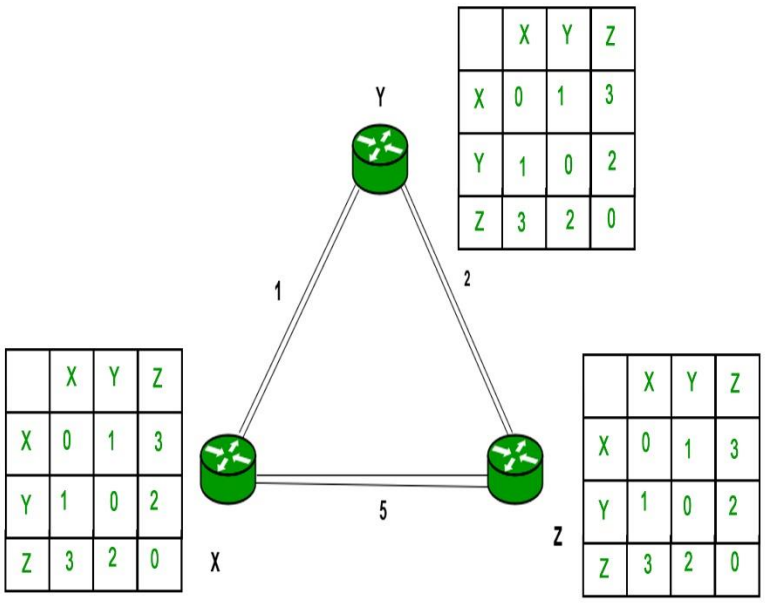


	X	Y	Z
X			
Y	1	0	2
Z			

	X	Y	Z
X	0	1	~~5~~ 3
Y	1	0	2
Z	~~1~~ 3	2	0

	X	Y	Z
X			
Y			
Z	5	2	0

Finally the routing table for all –



	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

Applications of Distance Vector Routing Algorithm

The Distance Vector Routing Algorithm has several uses:

- **Computer Networking**: It helps route data packets in networks.
- **Telephone Systems**: It's used in some telephone switching systems.
- **Military Applications**: It has been used to route missiles.

Advantages of Distance Vector routing

- **Shortest Path**: Distance Vector Routing finds the shortest path for data to travel in a network.
- **Usage**: It is used in local, metropolitan, and wide-area networks.
- **Easy Implementation**: The method is simple to set up and doesn't require many resources.
- **Quick Convergence**: It quickly adjusts to changes in the network, finding the best route to a destination fast.

Disadvantages of Distance Vector Routing Algorithm

- It is slower to converge than link state.

- It is at risk from the count-to-infinity problem.
- It creates more traffic than link state since a hop count change must be propagated to all routers and processed on each router. Hop count updates take place on a periodic basis, even if there are no changes in the network topology, so bandwidth-wasting broadcasts still occur.
- For larger networks, distance vector routing results in larger routing tables than link state since each router must know about all other routers. This can also lead to congestion on WAN links.

Hierarchical Routing

Hierarchical routing protocols (HRP) consist of a hierarchical topology to organize the network and routing information. Multiple layers and levels are introduced in a network. Each layer may be assigned a different responsibility like forwarding packets, maintaining routing tables, etc. HRPs are valuable for large networks, as they provide the capability of organizing network information and reducing the amount of routing information that should be exchanged between nodes. Hence, HRPs demonstrate significant scalability and fault tolerance. This is attributed to their hierarchical structure, which provides redundancy and facilitates the efficient distribution of routing data throughout the network.

Advantages of HRP

- **Scalability:** Hierarchical routing protocols exhibit excellent scalability by partitioning the network into smaller segments or areas. This division reduces the demand for routing tables and updates on each router, enhancing network efficiency and decreasing overall network traffic.
- **Better Traffic Control:** Hierarchical routing protocols demonstrate superior traffic management compared to flat routing protocols. The hierarchical framework enables more efficient traffic control, mitigating the need for unnecessary routing updates and preventing loops in the network.
- **Easy to Manage:** The organisational framework in these protocols facilitates simplified management and maintenance. Segmentation of the network into manageable sections enhances the ease of troubleshooting and diagnosing issues.

Disadvantages of HRP

- **Complexity**: Hierarchical routing protocols tend to be more intricate compared to flat routing protocols. The presence of additional layers and segments necessitates more extensive configuration, posing potential challenges in implementation.
- **Latency**: Latency may be introduced into the network due to the presence of additional layers and segments. Such delays in data transmission can pose challenges, particularly for real-time applications.

❖ **Congestion**

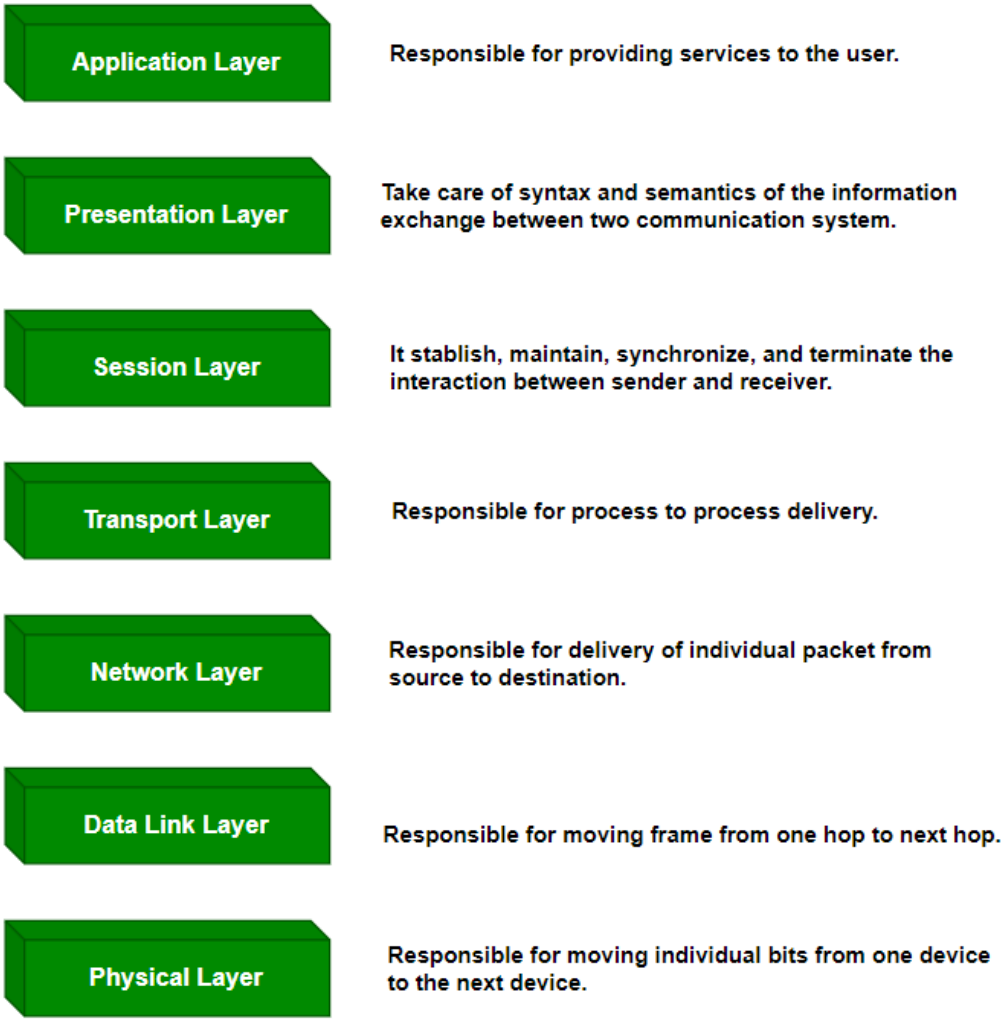
Congestion in a computer network happens when there is too much data being sent at the same time, causing the network to slow down. Just like traffic congestion on a busy road, network congestion leads to delays and sometimes data loss. When the network can't handle all the incoming data, it gets "clogged," making it difficult for information to travel smoothly from one place to another.

Congestion Control Algorithm

- Congestion Control is a mechanism that controls the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- **Congestive-avoidance algorithms** (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.
- There are two congestion control algorithms which are as follows:
  a) Leaky Bucket Algorithm
  b) Token Bucket Algorithm

❖ OSI Reference Model

The OSI model, created in 1984 by ISO, is a reference framework that explains the process of transmitting data between computers. It is divided into **seven layers that work together** to carry out specialised **network functions**, allowing for a more systematic approach to networking.



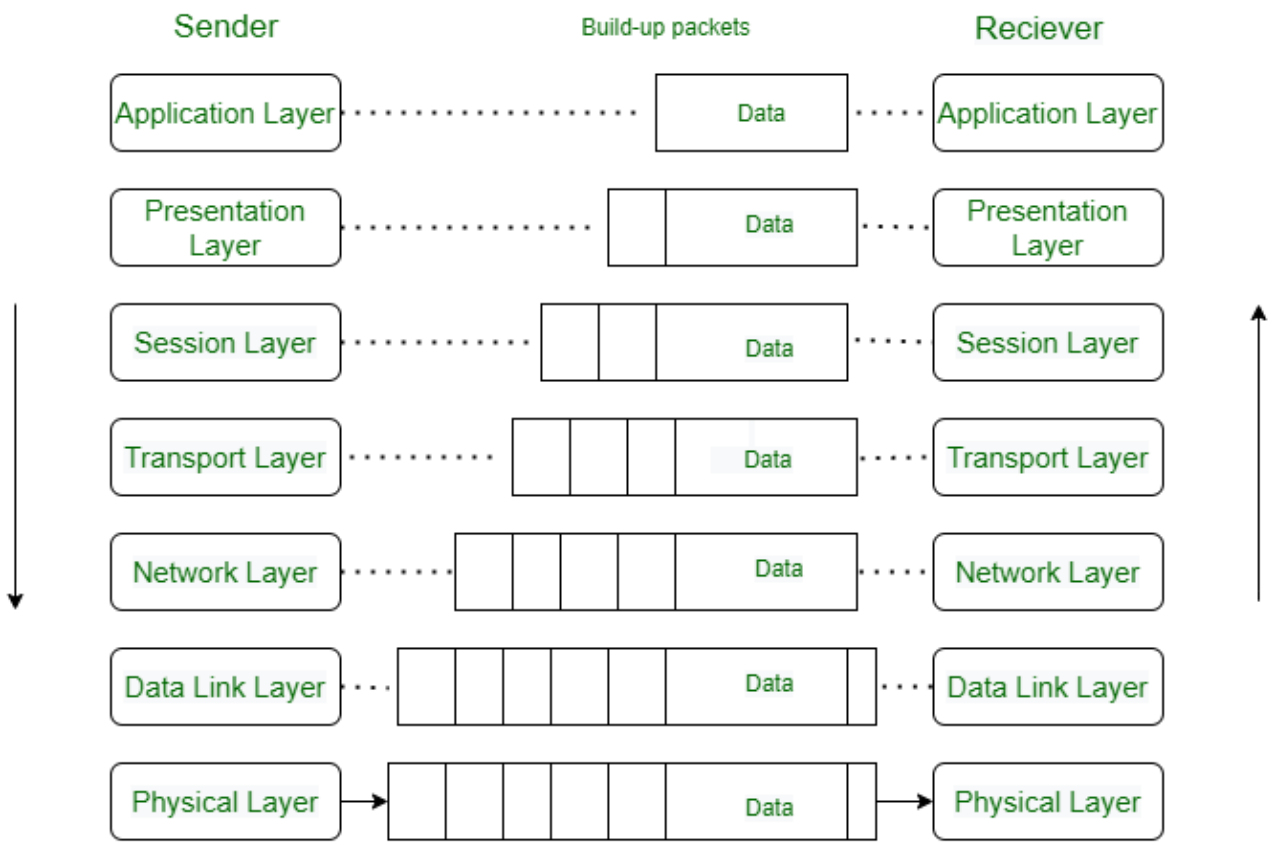
Data Flow in OSI Model

When we transfer information from one device to another, it travels through 7 layers of OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.

Data flows through the OSI model in a step-by-step process:

- **Application Layer:** Applications create the data.
- **Presentation Layer:** Data is formatted and encrypted.
- **Session Layer:** Connections are established and managed.
- **Transport Layer:** Data is broken into segments for reliable delivery.
- **Network Layer**: Segments are packaged into packets and routed.
- **Data Link Layer:** Packets are framed and sent to the next device.
- **Physical Layer:** Frames are converted into bits and transmitted physically.

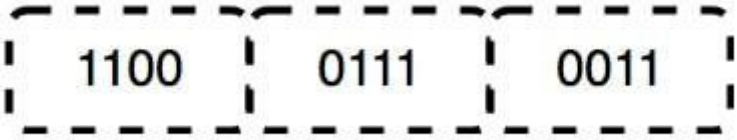
Each layer adds specific information to ensure the data reaches its destination correctly, and these steps are reversed upon arrival.



Let's look at it with an Example:
Luffy sends an e-mail to his friend Zoro.
Step 1: Luffy interacts with e-mail application like **Gmail**, **outlook**, etc. Writes his email to send. (This happens in **Layer 7: Application layer**)
Step 2: Mail application prepares for data transmission like encrypting data and formatting it for transmission. (This happens in **Layer 6: Presentation Layer**)
Step 3: There is a connection established between the sender and receiver on the internet. (This happens in **Layer 5: Session Layer**)
Step 4: Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information. (This happens in **Layer 4: Transport Layer**)
Step 5: Addressing of packets is done in order to find the best route for transfer. (This happens in **Layer 3: Network Layer**)
Step 6: Data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection. (This happens in **Layer 2: Data Link Layer**)
Step 7: Lastly Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.
After the email reaches the receiver i.e. Zoro, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Zoro's email client.
What Are The 7 Layers of The OSI Model?
The OSI model consists of seven abstraction layers arranged in a top-down order:
1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Physical Layer – Layer 1

The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of **bits.** It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.



Functions of the Physical Layer

- **Bit Synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit Rate Control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical Topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission Mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Note:

- *Hub, Repeater, Modem, and Cables are Physical Layer devices.*
- *Network Layer, Data Link Layer, and Physical Layer are also known as* ***Lower Layers*** *or* ***Hardware Layers****.*

Data Link Layer (DLL) – Layer 2

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.

The Data Link Layer is divided into two sublayers:

- Logical Link Control (LLC)
- Media Access Control (MAC)

The packet received from the Network layer is further divided into frames depending on the frame size of the NIC(Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.

The Receiver's MAC address is obtained by placing an ARP(Address Resolution Protocol) request onto the wire asking "Who has that IP address?" and the destination host will reply with its MAC address.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical Addressing:** After creating frames, the Data link layer adds physical addresses (**MAC addresses**) of the sender and/or receiver in the header of each frame.
- **Error Control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access Control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Note:
- *Packet in the Data Link layer is referred to as **Frame.***
- *Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.*
- *Switch & Bridge are Data Link Layer devices.*

Network Layer – Layer 3
The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.
Functions of the Network Layer
- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

Note:
- *Segment in the Network layer is referred to as **Packet**.*
- *Network layer is implemented by networking devices such as routers and switches.*

Transport Layer – Layer 4
The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as *Segments*. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.
At the sender's side: The transport layer receives the formatted data from the upper layers, performs **Segmentation**, and also implements **Flow and error control** to ensure proper data transmission. It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.

At the receiver's side: Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.
Functions of the Transport Layer
- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer
- Connection-Oriented Service
- Connectionless Service

1. Connection-Oriented Service: It is a three-phase process that includes:
- Connection Establishment
- Data Transfer
- Termination/disconnection

In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

3. **Connectionless service:** It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows

for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

Session Layer – Layer 5

This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Functions of the Session Layer

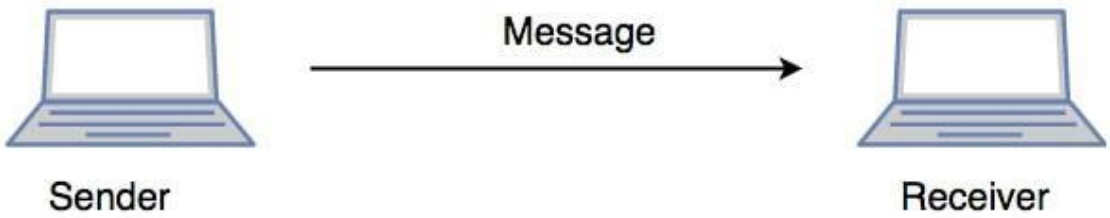
- **Session Establishment, Maintenance, and Termination:** The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Note:

- *All the below 3 layers(including Session Layer) are integrated as a single layer in the TCP/IP model as the "Application Layer".*
- *Implementation of these 3 layers is done by the network application itself. These are also known as **Upper Layers or Software Layers.***
- ***Device or Protocol Use :*** *NetBIOS, PPTP.*

Example

Let us consider a scenario where a user wants to send a message through some Messenger application running in their browser. The "**Messenger**" here acts as the application layer which provides the user with an interface to create the data. This message or so-called **Data** is compressed, optionally encrypted (if the data is sensitive), and converted into bits (0's and 1's) so that it can be transmitted.



Communication in Session Layer

Presentation Layer – Layer 6

The presentation layer is also called the **Translation layer**. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

Functions of the Presentation Layer

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Note: **Device or Protocol Use:** JPEG, MPEG, GIF.

Application Layer – Layer 7

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

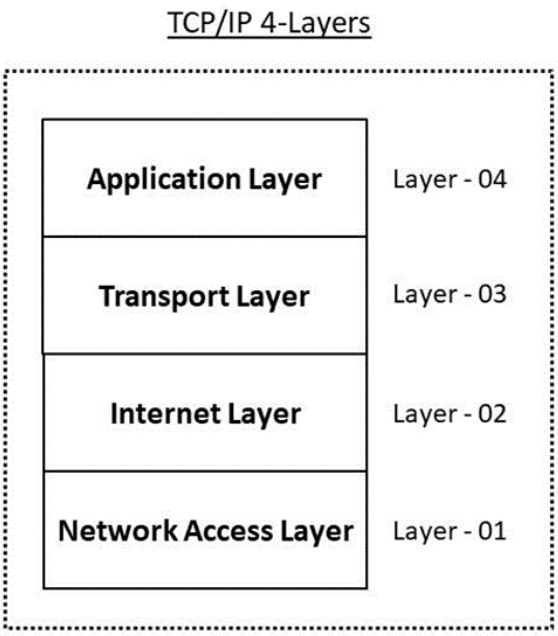
Example: Application – Browsers, Skype Messenger, etc.

Functions of the Application Layer
The main functions of the application layer are given below.

- **Network Virtual Terminal(NVT):** It allows a user to log on to a remote host.
- **File Transfer Access and Management(FTAM):** This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- **Mail Services:** Provide email service.
- **Directory Services:** This application provides distributed database sources and access for global information about various objects and services.

❖ **TCP/IP Model**

Whenever we want to send something over the internet using the TCP/IP Model, the TCP/IP Model divides the data into packets at the sender's end and the same packets have to be recombined at the receiver's end to form the same data, and this thing happens to maintain the accuracy of the data. TCP/IP model divides the data into a 4-layer procedure, where the data first go into this layer in one order and again in reverse order to get organized in the same way at the receiver's end.



1. Network Access Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and "framing" are also provided by this layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

2. Internet or Network Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

- **IP:** IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is

growing as the number of IPv4 addresses is limited in number when compared to the number of users.

- **ICMP:** ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:** ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

Example: Imagine that you are using a computer to send an email to a friend. When you click "send," the email is broken down into smaller packets of data, which are then sent to the Internet Layer for routing. The Internet Layer assigns an IP address to each packet and uses routing tables to determine the best route for the packet to take to reach its destination. The packet is then forwarded to the next hop on its route until it reaches its destination. When all of the packets have been delivered, your friend's computer can reassemble them into the original email message.

In this example, the Internet Layer plays a crucial role in delivering the email from your computer to your friend's computer. It uses IP addresses and routing tables to determine the best route for the packets to take, and it ensures that the packets are delivered to the correct destination. Without the Internet Layer, it would not be possible to send data across the Internet.

3. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:** Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

4. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

- **HTTP and HTTPS:** HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

The host-to-host layer is a layer in the OSI (Open Systems Interconnection) model that is responsible for providing communication between hosts (computers or other devices) on a network. It is also known as the transport layer.

Some common use cases for the host-to-host layer include:

- **Reliable Data Transfer:** The host-to-host layer ensures that data is transferred reliably between hosts by using techniques like error correction and flow control. For example, if a packet of data is lost during transmission, the host-to-host layer can request that the packet be retransmitted to ensure that all data is received correctly.
- **Segmentation and Reassembly:** The host-to-host layer is responsible for breaking up large blocks of data into smaller segments that can be transmitted over the network, and then reassembling the data at the destination. This allows data to be transmitted more efficiently and helps to avoid overloading the network.
- **Multiplexing and Demultiplexing:** The host-to-host layer is responsible for multiplexing data from multiple sources onto a single network connection, and then demultiplexing the data at the destination. This allows multiple devices to share the same network connection and helps to improve the utilization of the network.
- **End-to-End Communication:** The host-to-host layer provides a connection-oriented service that allows hosts to communicate with each other end-to-end, without the need for intermediate devices to be involved in the communication.

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